

Fair Orientations: Proportionality and Equitability

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- ① Background
- ② Preliminaries & Notions
- ③ Results
- ④ Conclusion & References

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Background: Fair Division with Relevance

The Real-World Constraint: Relevance

- In practice, items cannot be given to just *anyone*.
- **Goods:** A football match must be hosted by one of the two competing teams.
- **Chores:** Railway maintenance between City A and City B is only relevant to the authorities of A and B.

The Orientation Model

[CFKS23]: **Vertices** are agents, **Edges** are items.

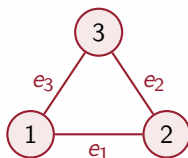
Allocating items means assigning a direction to each edge (**Orientation**).

No self-cycle is considered.

Modeling Relevance with Graphs

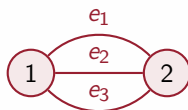
The topology defines how items can be allocated.

Simple Graph



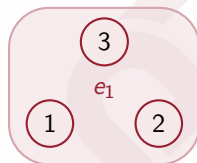
- $n_e = 2$ for all items.
- At most **one** item between any pair of agents.

Multigraph



- $n_e = 2$ for all items.
- **Multiple** items can exist between the same pair.

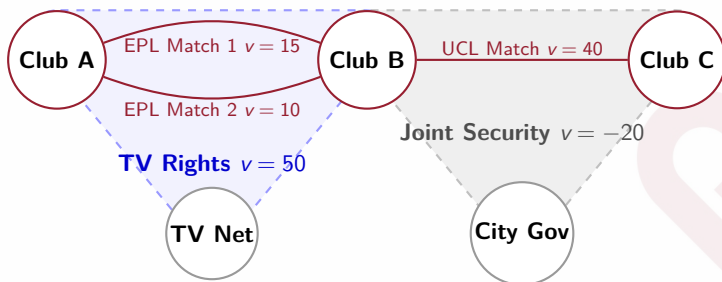
Hypergraph (General)



- $n_e \geq 1$ (any number).

Fair Division with Relevance

Fair division can be a massive, interconnected network of relevance.



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Formalizing the Model

- **Agents & Items:** A set of agents $N = \{1, \dots, n\}$ and items $E = \{e_1, \dots, e_m\}$.
- **Relevance Constraint:**
 - $E_i \subseteq E$: The set of items relevant to agent i .
 - $N_e = \{i \in N \mid e \in E_i\}$: The agents to whom item e is relevant.
- **Valuations (Utility):**
 - **Additive** utility: $v_i(S) = \sum_{e \in S} v_i(e)$ for any bundle $S \subseteq E$.
 - $v_i(e) \geq 0$ (**Good**), $v_i(e) \leq 0$ (**Chore**), and $v_i(e) = 0$ for $e \notin E_i$ (**Zero for irrelevant**).
- **Orientation:** $\pi = (\pi_1, \dots, \pi_n)$ where $\pi_i \subseteq E_i$ (**Relevant only**), $\bigcap_{i \in N} \pi_i = \emptyset$ (**No overlap**), and $\bigcup_{i \in N} \pi_i = E$ (**No remaining**).

Progress: Envy-free Orientations

For orientations, EF and EFX are known not to exist for all kinds of graphs. Only EF1 for goods is known to exist, even for hypergraphs. **What about EF1 for chores?**

Envy-Freeness Relaxations: EFX vs. EF1

For Goods ($v \geq 0$):

- **EFX:** $\forall e \in \pi_j$ s.t. $v_i(e) \geq 0$, $v_i(\pi_i) \geq v_i(\pi_j \setminus \{e\})$
- **EF1:** $\exists e \in \pi_j$ s.t. $v_i(e) \geq 0$, $v_i(\pi_i) \geq v_i(\pi_j \setminus \{e\})$

For Chores ($v \leq 0$):

- **EFX:** $\forall e \in \pi_i$ s.t. $v_i(e) \leq 0$, $v_i(\pi_i \setminus \{e\}) \geq v_i(\pi_j)$
- **EF1:** $\exists e \in \pi_i$ s.t. $v_i(e) \leq 0$, $v_i(\pi_i \setminus \{e\}) \geq v_i(\pi_j)$

Progress: Equitable Orientations

We disproved the existence of equitability in orientations.

Equitability Relaxations: EQX vs. EQ1

For Goods ($v \geq 0$):

- **EQX:** $\forall e \in \pi_j$ s.t. $v_j(e) \geq 0$, $v_i(\pi_i) \geq v_j(\pi_j \setminus \{e\})$
- **EQ1:** $\exists e \in \pi_j$ s.t. $v_j(e) \geq 0$, $v_i(\pi_i) \geq v_j(\pi_j \setminus \{e\})$

For Chores ($v \leq 0$):

- **EQX:** $\forall e \in \pi_i$ s.t. $v_i(e) \leq 0$, $v_i(\pi_i \setminus \{e\}) \geq v_j(\pi_j)$
- **EQ1:** $\exists e \in \pi_i$ s.t. $v_i(e) \leq 0$, $v_i(\pi_i \setminus \{e\}) \geq v_j(\pi_j)$

Progress: Redefine Proportionality for Fair Orientations

The $1/n$ Threshold Fails in Orientations

A chore is relevant only to Alice and Bob; there are 100 people.

- Since the chore must be taken either by Alice or Bob, Alice's reasonable expectation is to undertake $\frac{1}{2}v$, not $\frac{1}{100}v$.

We replace the global n with the item-specific $n_e = |N_e|$:

$$\text{PROP}_i = \sum_{e \in E_i} \frac{1}{n_e} v_i(e),$$

$n_e = 2$ for simple graphs and multigraphs.

An orientation is **PROP** if $v_i(\pi_i) \geq \text{PROP}_i$ for all $i \in N$.

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A Complete Landscape of Fair Orientations Results

Fairness	Pure Goods-Instance	Pure Chores-Instance
PROP	✗ + NP-c (Simple)	✗ + NP-c (Simple)
PROPX	✗ + NP-c (Simple)	✗ + NP-c (Simple)
PROP1	✓ + P + Fractional Pareto Optimal	(Hyper, Holds for Mixed Instances)
EQ	✗ + NP-c (Simple)	✗ + NP-c (Simple)
EQX	✗ + NP-c (Simple)	✗ + NP-c (Simple)
EQ1	✗ + NP-c (Simple)	✗ + NP-c (Simple)
EF	✗ + NP-c (Simple)	✗ + NP-c (Simple)
EFX	✗ + NP-c (Simple, [CFKS23])	✗ + NP-c (Multi, [Hsu25]) / ✗ + P (Simple, [ZWLL24; HK25])
		✗ + NP-c (Simple, Mixed Instances, [ZWLL24])
EF1	✓ + P (Hyper, [DEGK25])	✗ + NP-c (Multi) / ✗ + P (Simple)

- highlight our results.
- ✗ + NP-c = May not exist, NP-complete to decide;
 - ✗ + P = May not exist, Polynomial-time to decide;
 - ✓ + P = Always exist, Polynomial-time to compute.

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Conclusion

- EF1 orientations for chores in multigraphs $\implies$ NP-complete.
- EF1 orientations for chores in simple graphs $\implies$ Not Exist, Polynomial-time.
- PROP1 and fPO for mixed manna $\implies$ Always Exist, Polynomial-time.
- PROP and PROPX $\implies$ Not Exist, NP-complete.
- EQ, EQX, and EQ1 $\implies$ Not Exist, NP-complete.

Future: Do tighter notional boundaries between PROP1 and PROPX exist for orientations?

The End

Thank you for your listening!

Any questions?

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